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Four Decades of Digital Waveguides

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Digital waveguide physical modeling offers efficient simulation of acoustic wave propagation as compared to general finite-difference schemes commonly used in computational physics. This efficiency has enabled the real-time implementation of physically modeled musical instruments and sound effects, as well as real-time vocal models and artificial reverberation. This paper provides an overview of the historical evolution and applications of digital waveguide modeling and highlights recent advances in the field. Parametric optimization using classical, evolutionary, and neural approaches are also discussed and compared. Digital waveguides provide physically accurate simulations with reduced computational cost and can now be optimized with modern machine learning and differentiable digital signal processing techniques.

Keywords: digital filters, sound synthesis, optimization

1 INTRODUCTION

Since their introduction 40 years ago, *digital waveguide* (DWG) models have become a cornerstone of physically based sound synthesis and processing, spanning musical-instrument modeling, sound effects, artificial reverberation, and singing-voice synthesis [1–5]. Here, the authors review four decades of DWG research, drawing together their historical evolution, theoretical foundations, and modern applications.¹

In musical sound synthesis, physics-based equations are rarely used, even when the goal is to create realistic and expressive models of real-world instruments and effects. Instead, the prevailing approach often involves making numerous recordings that span the desired range of sounds—a process that is both labor-intensive and memory-

hungry. Historically, this preference arose because solving the *differential equations* of physics by means of finite-difference schemes was computationally prohibitive. Although advances in methods have made physical modeling more feasible, further acceleration remains a valuable objective.

The digital-waveguide approach to acoustic modeling tackles this challenge by optimizing the physical computations for nearly lossless 1D wave-propagation media, such as vibrating strings, acoustic tubes, and woodwind bores. In the frequency domain, all such systems exhibit quasi-harmonic resonant modes. The method has also been extended beyond one dimension through *waveguide meshes*. A rectangular lattice of 1D waveguides can approximate a 2D membrane, enabling efficient simulation of membranes and plates, with *N*-dimensional extensions following similarly.

The rest of this article is organized as follows. SEC. 2 charts key developments from the 1985 reverberator prototype to contemporary DWG-based products, comparing to related methods. SEC. 3 lays out the foundations and building blocks of DWGs. SEC. 4 surveys major extensions, and SEC. 5 surveys methods to optimize the DWG parameters, from classical physics-based filter-design and system-identification techniques to genetic, modern machine learning and differentiable digital signal processing (DDSP) techniques.

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¹Interactive audio examples and supplementary materials are available at <https://joshreiss.github.io/digital-waveguides-review/>. The full source code is openly accessible, and the demos synthesize audio in real time within any modern web browser using the Web Audio API.

2 HISTORICAL BACKGROUND

DWGs were introduced in the 1985 paper “A New Approach to Digital Reverberation Using Closed Waveguide Networks” [6]. The backstory is that Gary Kendall was working on large artificial reverberation systems, which he discussed with Julius Smith on the way to a conference. Kendall’s problem was that when he added a new feedback connection to enrich the response, the network would usually go unstable.

Since Schroeder and Logan in 1961 [7], there have been well-developed methods for using all-pass filters to walk the boundary between stable and unstable systems for artificial reverberation. But there was not yet a general approach for adding arbitrary signal feedback paths without significantly affecting stability or overall reverberation time. An all-pass construction approach was needed that supported any signal-path delay and interconnection topology. Smith pondered this while studying Belevitch [8] regarding all-pass networks, and the idea of closed networks of lossless DWGs struck him as a promising solution.

In hindsight, DWGs can be regarded as a straightforward application of the following:

1. D’Alembert’s *traveling-wave solution* of the wave equation [9];
2. *Scattering theory* implicit in d’Alembert’s derivation and later developed for transmission lines;
3. *Sampling* of bandlimited signals and systems as taught by Whittaker [10], Nyquist [11], and Shannon [10].

However, it was many years after d’Alembert’s solution that Lagrange [13] calculated the first demonstration of *reflection* and *transmission* at the boundary of two ideal strings having different mass densities (1759), and scattering theory developed about a century later in optics (e.g., Fresnel 1823 [14], Lord Rayleigh 1871 [15]). The concept of *wave impedance* awaited Heaviside’s “telegrapher’s equations” (coining “impedance” in 1886 [16]).

It can also be argued that standard physical-modeling formulations teach away from DWGs, because one normally begins with a differential equation giving constraints at every point in time and space, and that is discretized to make a generative model using *finite differences*. Since all practical models require *damping*, pure delay lines do not emerge from practical finite-difference schemes. One must come up with the *psychoacoustically valid approximation* of summarizing losses and dispersion at waveguide boundaries. Schroeder introduced delay-line simulation of sampled 1D wave propagation [17], extended by Moorer to include wall-reflection damping filters [18]. In the case of 1D linear, time-invariant waveguides, point-to-point transfer functions remain invariant as damping and/or dispersion filtering are *commuted* with delay elements to either endpoint, yielding an exact simplification; in higher dimensions, and for nonlinear or time-varying waveguides, it usually remains a high-quality approximation, despite the lack of commutative invariance.

In the early history of digital signal processing, there was a well-developed technology of ladder and lattice digital filters that could be derived from sampled traveling-wave theory [19]. Along those lines, a digital singing voice having a ladder-filter vocal tract was demonstrated at Bell Labs by Kelly and Lochbaum [20] with computer-music accompaniment by Mathews [21]. To get to a DWG voice model, it was only necessary to “erase” *scattering junctions* over sufficiently uniform stretches of vocal tract, such as from the larynx to the velum.

It is surprising that ladder filters were not routinely being used as computationally efficient and robust physical models of wave propagation. Instead, their use was normally abstracted away from their physical underpinnings, leaving single-input single-output digital filters, insufficient for use as physical modeling building blocks [22]. The field of signal processing became more rooted in applied math, abandoning its origins in physical analog circuits for the most part, with one exception being *wave digital filters* (WDFs; SEC. 2.3).

2.1 Digital Waveguide Synthesis

In the original DWG paper [6], it was suggested that “waveguide structures can provide accurate models of coupled vibrating strings, wind instruments, reed instruments, and many other physical systems. ... coupled to a nonlinear ‘excitation element,’ such as a reed, bow, switching air-jet, or lips” [23]. Subsequent work in that direction led to “Efficient Simulation of the Reed-Bore and Bow-String Mechanisms” [24] the following year.

Coincidentally, Stanford’s FM synthesis patent was nearing expiration, so the Stanford Office of Technology and Licensing sent the patent’s author David Lovejoy to the Center for Computer Research in Music and Acoustics to obtain all the latest papers. A patent application was submitted entitled “Digital Signal Processing Using Closed Waveguide Networks.” It was initially rejected due to its emphasis on “closed networks of lossless DWGs” (for reverberator prototypes) and the patent office’s rule that all proposed “perpetual-motion machines” lack utility because they are physically impossible to realize [25]. After a lot of back and forth, the patent was issued [26], so lossless reverberator prototypes apparently became the first patented virtual perpetual-motion invention in the United States (at least when roundoff-error feedback is used, which was done in some cases).

Yamaha began developing waveguide synthesis right away and in 1989 signed a nonexclusive license with Stanford. Yamaha’s Director of Engineering was present at the presentation of [24], as was Bob Moog, who wrote a newsletter article about it. In 1994, the Yamaha VL1 Virtual Lead synthesizer [27] was introduced, based on DWG modeling. The time was ripe for this approach to physical modeling synthesis, as computational power at the time could not afford a full finite-difference-scheme approach as would be typical in computational physics.

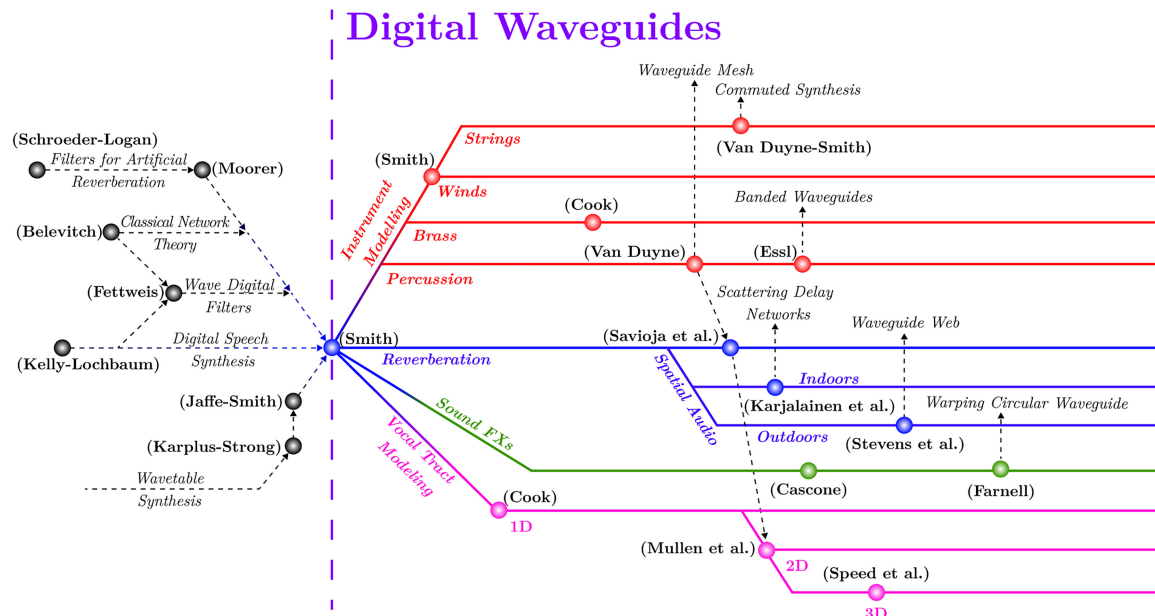


Fig. 1. The lineage of DWG modeling.

2.2 Relation to the Karplus-Strong Algorithm

The Karplus-Strong Algorithm (KSA) [28], introduced in 1983, is often described as physical modeling synthesis. But that is not how it was understood at the time. Instead, KSA was seen as *wavetable synthesis* that modified its wavetable on each pass through the table. For plucked string sounds, each sample was replaced by the average of itself and the previous sample, requiring no multiplies. Pitch control was achieved by setting the wavetable length to $N = f_s/f_0 - 1/2$ samples, where f_s and f_0 are the sampling and fundamental frequency respectively. As Kevin Karplus and Alex Strong noted [28]:

...interesting to compare the theoretical analysis of our synthesis technique [calculating overtone decay times] with an existing analysis of a guitar, lute, mandolin, or other string instrument. Unfortunately, we could not find a published analysis and did not have the tools to perform our own analysis. Some previous work has been done using physical models for synthesizing string sounds (Hiller and Ruiz 1971 [29]), but these models do not help to explain the high quality of the sound produced with our technique. ... interesting to convert the recurrence relations to differential equations, and to assign a physical interpretation.

Strong played in a string quartet with music doctoral student David Jaffe, to whom he demonstrated the “Digitar” algorithm. Jaffe brought it to electrical engineering Ph.D. student Smith to see about making it play in tune, adding dynamic level variations, and so on. Smith translated the KSA to a signal processing formulation and recast it as a delay line with explicit additive input, as seen in Fig. 2. It was then straightforward to develop Jaffe’s desired signal-processing extensions [30].

The tuning problem was solved by calculating the phase delay of the *attenuation filter* [31, p. 170] (now more general than the two-point average seen in Fig. 2) and compensat-

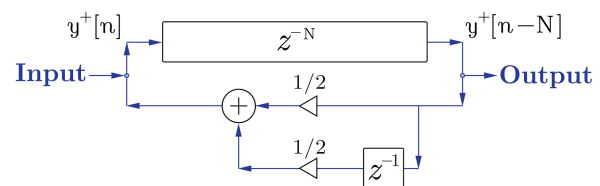


Fig. 2. A digital signal-processing diagram of the Karplus-Strong Digitar algorithm, with additive input added.

ing with delay-line interpolation—initially first-order all-pass interpolation [31, p. 175] and later Lagrange polynomial interpolation [32, 33]. Smith recognized the filter’s equivalence to point-to-point transfer-functions on an ideal string [31, pp. 158–194]. Impulse responses representing the vibrating-string “history” had already been introduced for bowed strings [34], but no one had suggested a round-trip string impulse-response as simple and sparse as that of a delay-line feeding back to its input through a two-point average. These insights gave rise to the pick-position illusion added in the Extended Karplus Strong algorithm [24]. Modulating the attenuation filter explicitly alters the decay and timbre. As will be seen in SEC. 5.2, multiple attenuation filter designs have been proposed.

This was not yet DWG modeling, however, because all signal paths involved *unidirectional* transfer functions with no notion of physical parameters such as string tension, mass, and/or damping. DWGs, in contrast, were conceptually *bidirectional delay lines* associated with a real, positive wave impedance calculated as the square root of string tension divided by mass density [6, 16]. To avoid terminology collisions with ladder and lattice digital filters, “delay line” here means two or more samples of delay. Typically, the final DWG model is simplified to reduce computational complexity while maintaining the physical interpretation. As an example, KSA was derived as a simplified physi-

cal model of a lossy vibrating string, randomly struck and plucked at each sample across its length [16]. That interpretation becomes more transparent when the order of the string and its resonator are swapped, or “commuted.” This is shown in SEC. 3.5, where KSA emerges as a simplified waveguide model.

2.3 Relation to Wave Digital Filters

While DWGs hold *sampled traveling waves*, WDFs hold *bilinearly transformed traveling waves*. Both use traveling-wave scattering theory, which has its origins in 19th-century optics (Fresnel 1823), acoustics (Lord Rayleigh 1871), and electrical engineering (Heaviside 1886). Lord Rayleigh is said to have first observed the proportionality of pressure and velocity in a traveling wave [35]. The scattering formulation for circuit-theory ports was introduced by Belevitch [36], and “*s* parameters” have been used since as optional port descriptions by microwave engineers, physicists, and others. This formulation was taken to the digital domain by Fettweis [37, 38] who coined the term WDF.

The *bilinear transform*, introduced by Tustin [39] and Steiglitz [40, 41], maps continuous voltage waves to discrete time in WDFs. Unlike sampling, bilinear transforms introduce no aliasing. The price for this is *frequency warping* according to $\omega_c \propto \tan(\omega_d T/2)$, where ω_c and ω_d denote continuous-time and discrete-time radian frequency, respectively, and T denotes the sampling interval in seconds. The WDF port impedances are chosen such that capacitors and inductors become unit delays [16], thus preserving system order.

Both DWGs and WDFs simulate traveling-wave components in known wave impedances. So they are readily interconnected despite having divergent frequency axes at high frequencies. WDFs are often used to model *lumped* systems involving masses, springs, and friction that are direct analogs of inductors, capacitors, and resistors. Examples include the *wave digital piano hammer* [42, 43] and the *wave digital tonehole model* [44]. WDFs are also used to create real-time digital models of nonlinear audio effects [45, 46] such as diode clippers [47] and tube amplifiers [48, 49]. DWGs efficiently implement distributed wave propagation media, such as vibrating strings, acoustic tubes, and tonal percussion. WDFs, in contrast, digitize lumped networks, such as mass-spring systems or resistor-inductor-capacitor circuits.

3 FOUNDATIONS

A DWG is a bidirectional delay line [16] associated with real, positive wave impedance R , which is given by the ratio of force to velocity in a traveling wave [16, 50, 51]. In practice, two separate delay lines may be used, one containing “left-going” traveling-wave components and the other “right-going” waves. Summing a signal into a DWG corresponds to superimposing a disturbance at one point. The original state is unaffected; the input signal enters the waveguide in superposition with the current traveling wave. This is depicted in Fig. 3, as a DWG with input at $n = 0$

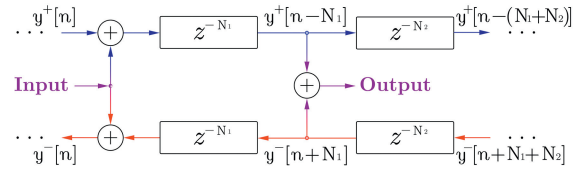


Fig. 3. Discrete-time simulation of ideal, lossless wave propagation using a DWG. Red lines indicate right-going traveling waves, while blue stands for left-going traveling waves.

and output at $n = N_1$.

The sum of left-going and right-going components yields the physical wave variable such as velocity, pressure, force, current, or voltage [16]. In many applications one can use a single delay line in which the first half corresponds to traveling waves to the right, say, while the second half contains left-going samples, thereby implicitly modeling a noninverting reflection at the midpoint of the single shared delay line.

A DWG models a 1D traveling-wave medium, such as an ideal vibrating string or acoustic tube. For frequencies up to the Nyquist limit, the modeling is exact in the ideal case because there are no losses, dispersion, or nonlinearities in an ideal string or acoustic tube.

3.1 Vibrating Strings

For vibrating strings, the simulated variables are typically velocity, force, displacement, or the acceleration at each sample point along the string. A typical choice is *transverse string velocity*, as measured directly by a magnetic pickup in an electric guitar. The scattering formulas are unchanged for displacement and acceleration, and force waves are similar with some sign flips.

Let $v[n, m]$ denote the physical velocity at temporal sample $n = 0, 1, 2, \dots$, and spatial sample $m = 0, 1, 2, \dots, M - 1$. So the digital string is simulated at M uniformly spaced samples over discrete time. For more physical time and position notation, $v_p[nT, mX] = v[n, m]$ can be defined, where T is the temporal sampling interval in seconds, and X is the spatial sampling interval in meters. Then in terms of traveling-wave components, there is [16]

$$\begin{aligned} v[n, m] &= v_+[n, m] + v_-[n, m] \\ &= v_+[n - m, 0] + v_-[n + m, 0] \\ &= v^+[n - m] + v^-[n + m], \end{aligned} \quad (1)$$

where $v_+[n, m]$ denotes the right-going and $v_-[n, m]$ the left-going traveling-wave components at time n and position m . The simpler $v^+[n - m]$ and $v^-[n + m]$ are defined, which only depend on time, making use of the fact that a spatial shift corresponds to a time delay or advance in a delay line.

Wave impedance of an ideal string is given by the square root of the string tension K times the *linear mass density* μ , or $R = \sqrt{K\mu}$ [16]. The wave impedance is thus the geometric mean of the two resistances to motion: a spring constant K and inertial mass μ . Similarly, *wave speed* is given by $c = \sqrt{K/\mu}$.

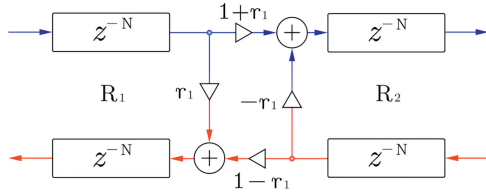


Fig. 4. Scattering junction.

Traveling velocity waves can be easily converted to traveling force waves using wave impedance R :

$$f^+[n] = R v^+[n], \quad (2)$$

$$[5pt] f^-[n] = -R v^-[n]. \quad (3)$$

Input to a DWG is usually added equally to left-going and right-going directions [16]. This is often accompanied by left and right reflections (“scattering”) at the input point, because physical inputs will change the wave impedance seen at that point in the waveguide [16].

3.2 Wave Impedance and Scattering

DWGs can be interconnected to form *digital waveguide networks* by joining them at scattering junctions. When a waveguide at impedance R_1 connects to a waveguide at a different impedance R_2 , scattering occurs, meaning partial reflection and transmission in both directions. Consider a right-going traveling-wave $v_1^+[n - M_1]$ at wave impedance R_1 encountering the impedance R_2 , which creates a scattering junction at the $R_1 : R_2$ interface, as shown in Fig. 4. The incident wave splits into a reflection $v_1^-[n - M_1]$ to the left and transmission $v_2^+(n)$ to the right. Given no second incoming wave from the right, $v_2^-(n) = 0$. Since all waves on the left at the junction have time index $n - M_1$, and all waves on the right at the junction have time index n , the indices can be omitted for notational simplicity. So v_1^+ , v_1^- and v_2^+ , and two constraints are present:

1. By Newton’s third law, the transverse forces must sum to zero at the junction:

$$[f_1^+ + f_1^-] + [f_2^+ + f_2^-] = 0 \quad (4)$$

$$\Rightarrow \frac{v_1^+ - v_1^-}{R_1} + \frac{v_2^+ - v_2^-}{R_2} = 0. \quad (5)$$

2. If the string does not break, then the string endpoint velocities are equal at the junction:

$$v_J = v_1^+ + v_1^- = v_2^+ + v_2^- = v_2^+. \quad (6)$$

These physical constraints are counterparts of Kirchhoff’s voltage and current laws in circuit analysis. The outgoing waves can be solved for:

$$v_1^- = v_1^+ \frac{R_1 - R_2}{R_1 + R_2}, \quad v_2^+ = v_1^+ \frac{2R_1}{R_1 + R_2}. \quad (7)$$

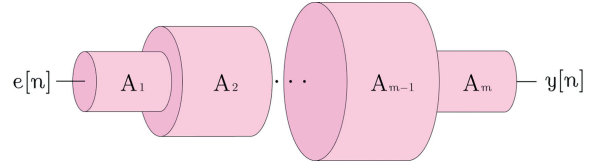


Fig. 5. Kelly-Lochbaum piecewise cylindrical tube model of the vocal tract, where “A” indicates the cross-sectional area of each corresponding section.

In terms of the *force reflection coefficient*

$$r_1 = \frac{R_2 - R_1}{R_1 + R_2}, \quad (8)$$

the complete *scattering equations* for this *series* junction of Fig. 4 can be written as

$$f_1^- = r_1 f_1^+, \quad f_2^+ = [1 + r_1] f_1^+, \quad (9)$$

$$v_1^- = -r_1 v_1^+, \quad v_2^+ = [1 - r_1] v_1^+. \quad (10)$$

It is a “series” junction because there is a common velocity, while forces sum to zero. By contrast, a “parallel” junction has a common force, and velocities sum to zero.

As $R_2 \rightarrow \infty$ (rigid termination), $v_1^- = -v_1^+$ (an inverting reflection), and $v_2^+ = 0$ (no transmission). Similarly, when $R_2 = 0$ (a “free end,” which can be imagined as a frictionless guide-pole), $v_1^- = v_1^+$ (noninverting reflection) and $v_2^+ = 2v_1^+$ (giving velocity continuity across the junction) are obtained. Finally, when $R_2 = R_1$, $v_1^- = 0$ and $v_2^+ = v_1^+$ (no scattering) are obtained.

Note that power is conserved (i.e., the scattering is lossless). The incident traveling power is $f_1^+ v_1^+$, and the outgoing power after scattering is $f_2^+ v_2^+ - f_1^- v_1^-$. Calculation is easy in terms of the reflection coefficient:

$$\begin{aligned} f_2^+ v_2^+ - f_1^- v_1^- &= [1 + r_1] f_1^+ [1 - r_1] v_1^+ \\ &\quad - r_1 f_1^+ [-r_1 v_1^+] \\ &= [1 - r_1^2] f_1^+ v_1^+ + r_1^2 f_1^+ v_1^+ \\ &= f_1^+ v_1^+. \end{aligned} \quad (11)$$

3.3 Acoustic Tubes

The vocal tract in speech is often modeled as a piecewise-cylindrical acoustic tube [16], as shown in Fig. 5. In this model, the joining planes between different sections are scattering junctions (as in Fig. 4) in the sampled traveling-wave formulation.

Signal samples are typically acoustic *pressure*, and wave impedance in the m th tube section (pressure divided by volume-velocity) is given by $R_m = \rho c / A_m$, where ρ is air mass density (kilograms per cubic meter), c is the speed of sound (meters per second), and A_m is the cross-sectional area of the m th tube section (square meters).

The scattering formulas at the junction of cylindrical segments m and $m + 1$ can be written in terms of the reflection coefficient r_m , defined by

$$r_m = \frac{R_{m+1} - R_m}{R_{m+1} + R_m} = \frac{A_m - A_{m+1}}{A_m + A_{m+1}}. \quad (12)$$

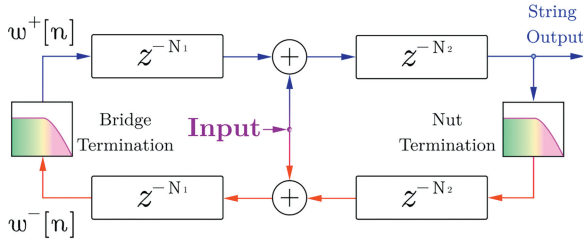


Fig. 6. Discrete simulation of a string using filters as bridge and nut terminations. In the rigid case, the filters become a -1 multiplication for velocity or displacement waves.

Thus, the reflection coefficient r_m depends on the *area ratio* $A_m/A_{m+1} = R_{m+1}/R_m$ between sections. Using this notation for the string gives $v_1^- = -r_1 v_1^+$, where the minus sign arises because velocity was used instead of pressure as the wave variable.

3.4 Waveguide String Models

A basic waveguide string model is based on a pair of delay lines, where the upper delay line simulates the wave propagation from the bridge to the nut and the lower delay line propagates in the opposite direction [16]. All samples move one position each sampling interval by incrementing (or decrementing) a pointer to a string endpoint in the circular-buffer implementation of the delay line. At each end of the string, the sampled wave reflects backward, optionally transmitting as well. In the lossless case, the reflection coefficient is -1 for velocity (or displacement) at a rigid termination. Ideal reflection at the bridge end means that the incoming velocity sample arriving along the lower rail gets its sign inverted as it jumps to the upper rail, where it travels toward the other end. However, oscillations never decay in this undamped model, so a periodic signal is produced after any excitation.

A more realistic waveguide string model has lowpass filters at one or both ends, which model the frequency-dependent propagation losses along the string as well as the reflection from the string end [52–54]. If the total gain of the round-trip filtering is less than 1 at all frequencies, the sample amplitudes decay gradually, which corresponds to the decay of the string vibration.

3.5 Commuted Synthesis

Since basic waveguide string models consist of linear, time-invariant elements, the elements can be reordered in series without changing the transfer function. So the two delay lines of a DWG may be combined into one doubly long delay line. Similarly, the two reflection filters can be commuted to the same point in the delay-loop and combined in a single filter realizing all the filtering necessary in the whole delay loop. This paper refers to this generic structure, similar to that of the KSA (see SEC. 2.2), as a *filtered delay loop* (FDL) [22, 55].

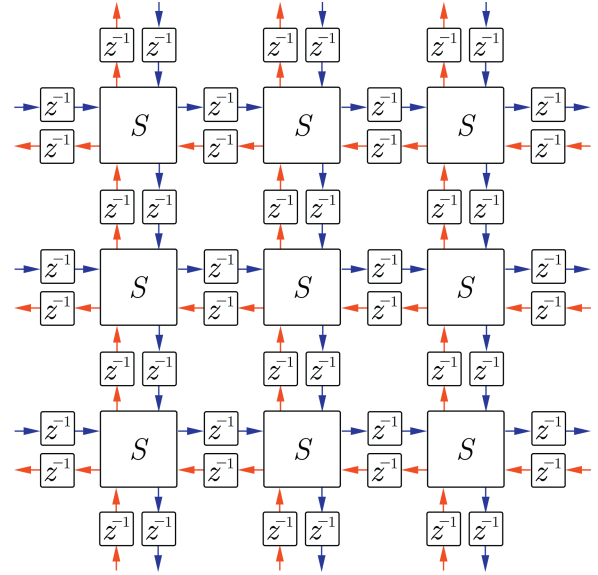


Fig. 7. 2D rectilinear waveguide mesh. S stands for scattering junctions.

In a plucked-string instrument, the output signal $y[n]$ can be expressed as a cascade of convolutions:

$$y[n] = e[n] * h[n] * b[n], \quad (13)$$

where $e[n]$ is the *excitation signal* (e.g., the pluck) and $h[n]$ and $b[n]$ stand for the impulse response of the string and instrument body, respectively. Since convolution is commutative, these elements can be reordered without changing the output.

This affords great computational savings for strings attached to resonators [56]. Instead of plucking a string and driving a high-order body resonator, such as a guitar body or piano soundboard [57, 58], one can pluck the resonator and feed that into the string [52, 56, 59]. The savings comes from storing only the impulse response (or “pluck response”) of the resonator $e[n] * b[n]$. So the large digital resonator models needed for acoustic stringed instruments (hundreds or thousands of poles and zeros occurring in the audio range) are replaced by a basic wavetable lookup and playback through a small filter that can model variations in plucking details. This also applies to bowed strings modeled as periodically plucked strings [56, 60, 61].

3.6 Digital Waveguide Mesh

The *digital waveguide mesh* (DWM) simulates a membrane or volume using a lattice of 1D strings, analogous in 2D to a tennis racket [62]. A 2D membrane is thus approximated by a mesh of intersecting 1D waveguides, as depicted in Fig. 7. Each intersection forms a four-port scattering junction. Typically, all wave impedances are equal so that the junction velocity v_J can be computed using no multiplies as $v_J = \frac{1}{2} \sum_{i=1}^4 v_i^+$ (i.e., sum of four incoming waves v_i^+ divided by 2). Each outgoing wave is then $v_i^- = v_J - v_i^+$ for a total of seven additions and one bit-shift (in fixed-point). Lossless wave propagation is exact along the diagonals but *dispersive* along horizontal and

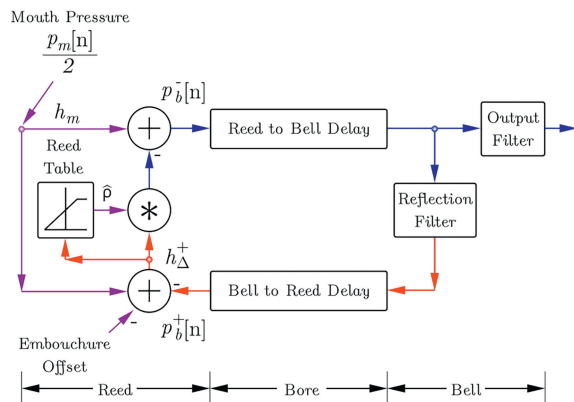


Fig. 8. DWG model of a single-reed woodwind instrument.

vertical directions [63]. To reduce dispersion at the cost of multiplies, the *triangular mesh* is a good choice [64–66].

For multiply-free scattering in 3D, the *tetrahedral mesh* geometry is preferred [67, 68]. Later mesh developments include acoustic room simulation [69–73], the bilinearly deinterpolated waveguide mesh [74, 75], high-frequency mode modeling [76, 77], the frequency-warped waveguide mesh [65, 78, 79], violin body modeling [80], boundary conditions [81], quadratic residue diffusers [82], extensions to higher dimensions [83], rigid and diffuse boundary modeling [84, 85], and vocal-tract modeling [86–90].

4 ADVANCEMENTS

4.1 Instrument Modeling

4.1.1 Winds

The first waveguide synthesis instrument was the clarinet [24] (Fig. 8). Pressure waves p propagate through the bore and interact with the mouthpiece, where subscripts m and b denote mouth and bore, and superscripts $+$ and $-$ indicate right-going and left-going waves. Half-pressures $h = p/2$ are used, where h_{Δ}^{+} across the reed corresponds to the pressure drop driving nonlinear flow. The clarinet reed was modeled as a massless spring flap [91, 92], acting as a nonlinear reflection coefficient $\hat{\rho}$ at the tube end—mapping the pressure difference across the reed to the resulting airflow—and a nonlinear variable gate for airflow into the tube.

The initial implementation of the reed-bore reflection-coefficient (a function of the incoming pressure waves from both sides, and the player's embouchure) was a linearly interpolated table-lookup. This presolved nonlinear scattering junction can be regarded as a 1D version of the *K Method* [93], also applicable to root-node scattering in nonlinear WDFs [45]. Later implementations included piecewise polynomial spline approximations [94, 95]. Some reed models also included mass [91, 96], which cannot be presolved, so a few Newton iterations or the like are used to solve for the nonlinear junction outputs.

The second waveguide wind-instrument model was the flute [97–100]. The mouthpiece excitation was modeled as an air jet [23] that swung in and out of the tube based on air-

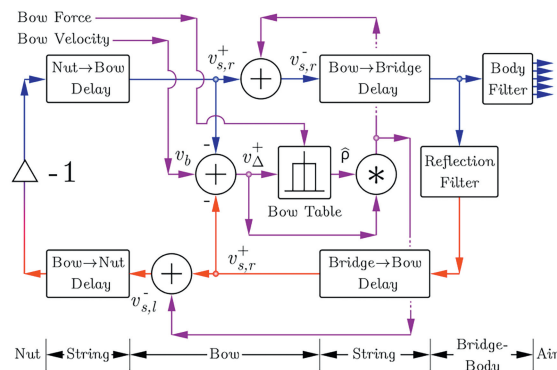


Fig. 9. DWG model of a bowed string.

jet dynamics and the traveling pressure wave in the bore at the mouthpiece. Since the flute mouthpiece is in the interior of the bore, two waveguides are used, one on each side of the excitation.

The first real-time waveguide saxophone was in the Yamaha VL-1 synthesizer [27]. It is said to have used Benade's two-cylinder model for the conical bore [101–103]. Conical bores become interesting near the tip, which behaves like an open end for pressure [104]. Another oddity is that the impulse-response of an increase in the conical taper-angle has a growing exponential component [105]. Woodwind toneholes were modeled as variable scattering junctions in the DWG bore [106, 107], and a wave digital tonehole was developed [44].

4.1.2 Strings

Waveguide synthesis for strings began with simplified bowed strings [24], which affixed a bow-string interaction model [23, 34] to a one-branch DWG network [6], as shown in Fig. 9. In the model, velocity-wave components v travel along the string, with subscripts s, r and s, l referring to string segments to the right and left of the bow, respectively. This divided the string into two waveguides joined by a nonlinear two-port scattering junction, where the bow-string interaction is governed by the differential velocity v_{Δ}^{+} (bow velocity minus incoming string velocity) and the nonlinear reflection function $\hat{\rho}$, which depends on bow force and velocity.

Bow position is varied by adjusting the relative lengths of the left and right delay lines, with interpolation providing continuous control. Assuming the bow-hairs present only a nonlinear frictional force (i.e., neglecting bow-hair traveling waves), a table-lookup or piecewise polynomial solution is obtained for the instantaneous reflection and transmission coefficients [24]. Later models introduced friction-loop hysteresis and thermodynamic models of rosin friction [108–112].

DWGs have been used for modeling various plucked string instruments, such as the acoustic guitar [54, 59, 113], Finnish kantele [114, 115], and geomungo, a traditional Korean instrument [116]. To capture the unusually wide vibrato effect of the geomungo, DWG synthesis with a time-varying attenuation filter was used [116]. Some key-

board instruments, which are also based on vibrating steel strings, have been emulated with DWGs, such the piano [51, 117–119], harpsichord [120], and Clavinet [121]. The electric guitar is a popular target for DWG synthesis [122–125].

4.1.3 Brasses

Waveguide brass instruments began with the trombone [94]. In brass models, the lips are typically modeled as one or two mass-spring elements driving the waveguide tube. Flared bells inspired applications of Sturm-Liouville theory [126], and the most efficient known filter structure for the flared bell is truncated infinite-impulse-response (IIR) filters [105].

4.1.4 Percussion

Tonal percussion, such as glockenspiel and triangle, are straightforward to model as DWGs, being composed of ≈ 1 D metal rods. Membrane percussion, such as tympani and other drums, can use a 2D waveguide mesh for their membranes, and a 3D mesh for the enclosed air cavity. Another approach to membranes, which has been applied to cymbals, is the *banded waveguides* technique, which identifies *closed wavetrains* modeled by 1D DWG loops [127–129].

4.1.5 The Synthesis Tool Kit and Faust

The Synthesis Tool Kit (STK) [95] is a major software repository of example DWG instruments. An extended subset of the STK examples was ported to the Faust distribution as well [130]. The Web Audio examples for this review include ports of STK examples.

In addition to ports of the original STK classes, the Faust Libraries [131] provide many additional synthesis and effects algorithms using DWGs. The term “waveguide” appears 55 times, as of this writing, spanning the libraries for oscillators, noises, filters, physical models, and miscellaneous effects.

4.2 Vocal Tract Modeling

The Kelly-Lochbaum model [20] is thought to be the first digital scattering-based model of the vocal tract, and it influenced ladder and lattice filter development for linear prediction of speech [19]. Smith has provided a detailed derivation of wave impedance and scattering theory starting with d’Alembert’s wave equation [22].

DWG modeling of *singing human voice*, introduced by Cook [132], can be considered a generalization and extension of the Kelly-Lochbaum model. A real-time interactive singing-voice model called SPASM (“Singing Physical Articulatory Synthesis Model”) [2, 133] included a nasal tract branching off from the vocal tract through a variable three way velum/pharynx junction, turbulence noise injectable at any point along the vocal tract, variable glottal and lip reflection coefficients, and separate outputs for neck, lip, and nose radiation. A related interactive implementation called “Pink Trombone” is available online [134].

Later advancements used DWMs (SEC. 3.6) for the vocal tract. Early 1D versus 2D comparisons and bandwidth con-

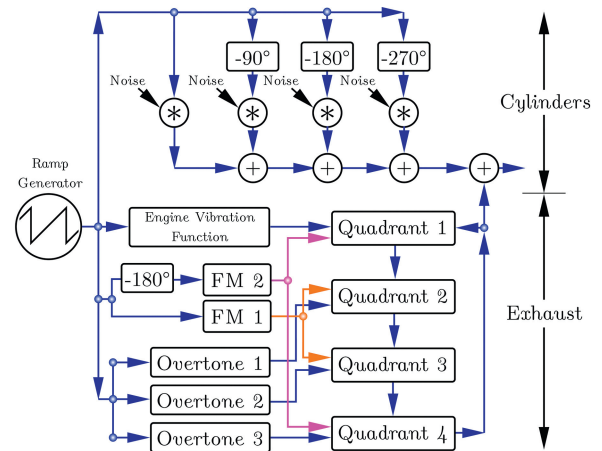


Fig. 10. Four-cylinder engine model using DWGs. Pink and orange arrows indicate FM control signals (not audio inputs).

trol [135, 136] revealed that 2D relaxation of the plane-wave assumption introduces lateral modes and provides approximately linear formant-bandwidth control across vowels [137]. Real-time dynamic articulations in a fixed rectangular 2D mesh using impedance mapping were then introduced [138], and 3D meshes that match measured transfer functions for tract analogs demonstrated close acoustic match to physical analogs over a wide frequency band [87, 139]. Recent results use magnetic resonance imaging-derived geometry and time-varying impedance maps to synthesize diphthongs with improved formant accuracy and naturalness [140] and voiced stop consonants via controlled occlusions in a 3D DWM [90].

Beyond the human laryngeal tract, waveguide principles have also been applied to other vocal organs. In songbirds, the syrinx—a bifurcated airway with vibrating membranes—has been modeled using classical waveguide synthesis [141–143]. Mammalian vocal tracts were implemented using a waveguide filter model in [144]. Subjective tests showed that it was effective in reproducing harsh, spectrally dense sounds such as a lion’s roar or a wolf’s growl.

4.3 Sound Effects

DWGs are also used to simulate various mechanical devices, thus creating realistic sound effects for use in creative media and computational modeling. The most common example of this is in simulating vehicle engine sounds, where DWGs are often used for engine cylinders and exhaust pipes. The first example was perhaps given by Cascone et al. [145], for which a Faust implementation is available [146].

Farnell introduced DWGs with antiphase frequency-modulated delays for simulating engine exhausts, controlled by a ramp generator mimicking mechanical inertia [4]. These delay lines are split into four quadrants, with some expanding and others compressing while maintaining constant length, as shown in Fig. 10. He coined this the *warping circular waveguide* [4]. The Engine Vibration Function applies a parabolic waveshaper to model low-

frequency body resonance, while *overtone generators* inject bursts of higher-frequency content at specific positions, simulating mechanical transients (valve clicks, combustion noise). Separately, cylinders are simulated by producing pulses from the phase-shifted ramp, multiplied by noise for jitter. This approach was further developed by Baldan et al. who described car engine sound synthesis with multiple DWGs [147]. Two subsequent works incorporated Baldan's implementation into soundscape synthesis of traffic sounds [148, 149].

Farnell's use of DWGs, though informal and often heuristic, was wide in scope. DWGs were incorporated into models of not just automotive engines, but models of old telephones, impacts on windows, switches and levers, and helicopters [4].

Other applications of DWGs for generating sound effects are of note. Oksanen et al. modeled the sound of a jackhammer using parallel waveguides for longitudinal and transversal vibrations [150]. Use of banded DWGs for generating the "twang" sound of a ruler being pushed while it is hung over a desk was described by Selfridge et al. [151], though they opted for a modal synthesis approach in their implementation.

In [152] and references therein, Keenan and Pauletto simulated the sound of an acoustic wind machine, which is a mechanical device that in turn simulates wind sounds. The machine has a slatted cylinder whose slats rub against a cloth as the cylinder moves. The resultant friction produces a wind-like sound. A DWG in series with low-pass and all-pass filters was used to simulate dispersion of the friction sound through one side of the cloth. Use of DWGs for real-time, controllable sound effect generation is discussed and put in the wider context of *procedural audio* in a review paper by Menexopoulos et al. [153].

4.4 Reverberation

The first *digital audio effect* using a 1D waveguide network was an "invertible waveguide reverberator," showcased in the finale of "Invisible Cities" ("City of Reflection" movement) by McNabb [154]. In this striking effect, a saxophone passage is scattered to "sonic dust" by a closed, lossless network, and then that network runs in reverse to congeal the dust back to the original sax passage. Despite this early artistic success, there was little widespread application to reverberation until the introduction of the waveguide mesh (see SEC. 3.6). Widely adopted feedback delay networks [155] can be described as DWGs having one scattering junction, but they were not derived that way.

Physical modeling networks of 1D waveguides for spatial audio awaited the advent of *scattering delay networks* (SDNs) [156, 157], illustrated in Fig. 11. In the SDN, the source drives the wall nodes via unmodified, unidirectional paths, while the receiver collects scattered components via unidirectional paths. The direct sound from source to receiver follows the free-field direct path. The first-order reflection paths from source to receiver pass through the scattering junctions. Four scattering junctions denoted S are located at the physical first-order-reflection points along

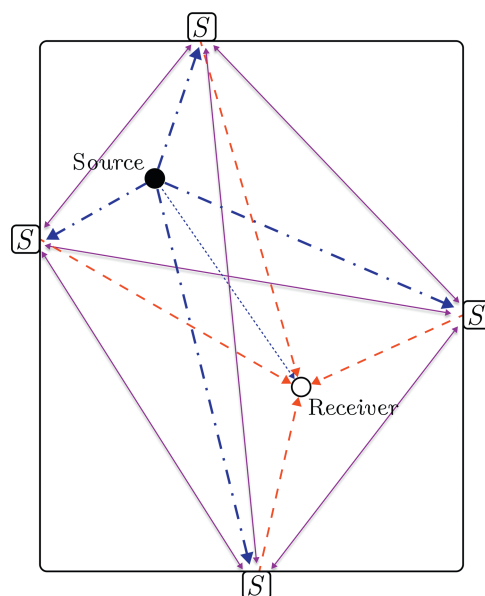


Fig. 11. SDN with wall nodes (S), bidirectional delay lines (purple), source paths (dash-dotted blue), receiver paths (dashed red), and direct path (dotted blue).

the walls. Six bidirectional delay lines interconnect the four scattering junctions, forming a digital waveguide network. The purpose of the network is to simulate the higher-order reflections efficiently, albeit with approximation error: while the scattering junctions are located correctly for first-order reflections, they are not generally correct for higher-order reflections.

Thus, the efficiency of the SDN reverberator is obtained by moving the point of reflection for each higher-order reflection from its exact location along the wall to the location of the first-order reflection on that wall. The main error is that the exact timings of the higher-order reflections are perturbed, but the echo distribution and density are preserved. As established by De Sena et al., "according to various objective measures of perceptual features, SDN achieves a reverberation quality similar to that of the [theoretically precise Image Method] while having a computational load one to two orders of magnitude lower" [157].

While SDNs target enclosed room acoustics, similar principles have been adapted for outdoor scenarios. The *Waveguide Web* is a digital reverberation algorithm designed to simulate the acoustics of sparsely reflecting outdoor environments, such as a scene with a few buildings, by using interconnected DWGs and scattering junctions to model reflection points [158]. The Waveguide Web extends the Treeverb architecture proposed by Spratt and Abel, which models the acoustics formed by trees in a forest [159]. The Treeverb, shown in Fig. 12, may be interpreted as a 2.5D model of a forest, where the source excites the tree nodes and the receiver gathers the postscattering components, including the horizontal acoustic path from the source to the receiver point and all scattering paths between the tree nodes. As more trees are added into the system, it becomes computationally inefficient, however.

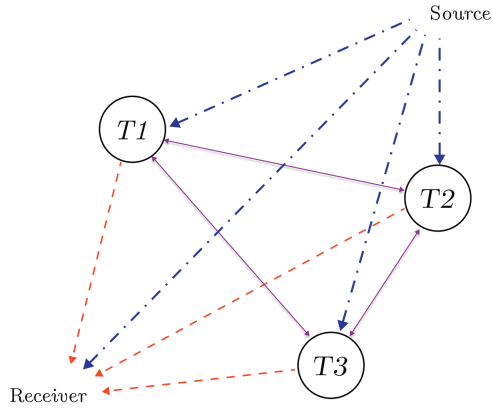


Fig. 12. Treeverb (DWG web). Tree-nodes (T_1 – T_3), bidirectional lines (purple), source paths (blue), and receiver paths (red).

In the Waveguide Web model proposed by Stevens et al. [158], each node combines the acoustic properties of an entity, such as a building, by storing its reflection properties for the first-order reflection of the source signal and the second-order reflections from the other nodes. For open scenes, such as a courtyard surrounded by a couple of facades, the necessary filters can be measured or simulated. This way, a fairly complicated scene can be simulated with a small number of nodes. The Waveguide Web model accurately characterizes both the first-order and the second-order reflections from the nodes [158]. Now the scattering junctions cannot be energy-preserving, since some of the acoustic energy escapes the scene between the buildings. A sky node may be defined to simulate such losses [158]. Nodes for ground reflections could also be incorporated, if they appear to be acoustically important.

5 PARAMETRIC OPTIMIZATION IN DIGITAL WAVEGUIDES

Often professionals wish to calibrate a DWG model to resemble a target sound for realism or to obtain physical parameters of the sound-producing object, such as bow force or reed pulse. In this sense, parametric optimization in DWGs can be viewed as an architecture-specific instance of the broader synthesizer sound-matching problem overviewed by Shier [168].

Over the last four decades, multiple strategies have been applied to DWGs, which are organized chronologically. *Physics-based* approaches rely on analytical relations and measurements. *Filter-design* methods approximate the DWG by placing poles, zeros, or coefficients of its transfer function $H(z)$. *System identification* methods precompute input-output mappings for online inference (for an early review of filter design and system identification, see [31]). *Genetic* approaches perform random searches where successful parameter sets survive and others are discarded. Finally, *neural* methods optimize parameters by backpropagating loss-gradients through differentiable neural operations. Neural approaches can be further divided into *black-box* methods, where the synthesizer's in-

ner workings are unknown, and *white-box* methods, where they are known and explicitly modeled. While the former imposes knowledge about the signal or system into the optimization method, the latter relies on learned mappings with minimal assumptions.

5.1 Physics-Based

Resonating acoustic bodies are often modeled via *modal expansion* to obtain an impulse response in the form of a sum of exponentially damped sinusoids,

$$x(t) \approx \sum_{n=1}^N A_n e^{-\alpha_n t} \cos(\omega_n t + \phi_n), \quad (14)$$

and the modal parameter set (A_n , ω_n , α_n) is estimated with some spectral method to design the attenuation filter of an FDL system. Guillemain et al. [160] and Ystad [161] match (ω_n , α_n) to viscothermal losses in tubes and stiffness-induced dispersion on strings. However, they report this method does not guarantee realism [160, 161]. In practice, physics-based approaches are often hybridized with other methods due to their potential to simplify optimization by fixing variables: Smyth and Abel, for example, anchor a clarinet waveguide with measured bell reflection filter to obtain its exciting reed pulse [162, 163]. As will be seen, the literature usually restricts filter design approaches to 1D structures, such as string or tube models, collapsed into FDLs (see SEC. 3.5).

5.2 Filter Design

Alternatively, (A_n , ω_n , α_n) can be used to derive the location of poles and zeros of the transfer function $H(z)$ through some filter-design method [53, 165]. The strategies used are summarized in Table 1. Fundamental frequency f_0 can be recovered as the lowest detected harmonic [166] or via harmonic spacing [167]; while many works report accurate estimation through autocorrelation [52, 53, 165, 168].

Instrument bodies can be simulated once the string $h[n]$ has been optimized. Välimäki et al. rearrange the terms of Eq. (13) to $y[n] * h^{-1}[n] = e[n] * b[n]$, obtaining the pluck response. This is then stored and reproduced online to simulate guitar, mandolin, kantele, and banjo [52]. Later, Tolonen and Välimäki note the nonharmonic modes of $b[n]$ (e.g., 100-Hz *Helmholtz resonance* in guitars) can bias the filter-design method of $h[n]$ and propose modeling $b[n]$ separately with warped-IIRs, which can be shared along multiple strings [52, 53, 165].

5.3 System Identification

Kestian and Smyth [169] use a Kelly–Lochbaum model (see Fig. 5) to precompute a lookup table, mapping area sections to formants. At runtime, nearest-neighbor matches the area from recordings [169]. Similarly, Smyth and Wang [170] estimate saxophone fingering with a 1D reed-bore DWG. The bore values are precomputed per fingering, choosing the one that maximizes time-domain cosine similarity from bell-mic recordings [170]. Serafin et al. use linear predictive coding and principal component analysis to extract features from a precomputed bowed-string (see

Table 1. Optimization methods for DWGs.

Paradigm	DWG	References	Method
Physics-based	Filtered delay loop	[160, 161]	Analytical relations
	Clarinet	[162, 163]	Real-life measurements
		[166]	Prony
Filter design	Filtered delay loop	[52, 53, 165, 210]	Iterative weighted error
		[118, 211, 212]	Steiglitz–McBride
System identification	Kelly–Lochbaum	[169]	Nearest neighbor
	Reed–Bore	[170]	Cosine similarity
	Bowed string	[171]	Gaussian mixture model
	Avian vocal tract	[143]	Maximum likelihood
	Flute	[99]	Spectral fitness
Genetic	Filtered delay loop	[172]	
	2D vocal tract DWM	[173]	Perceptually informed fitness
	Pink Trombone	[174, 175]	Covariance matrix adaptation evolution strategy
	String	[176]	Distal teacher
	Filtered delay loop	[177]	Neural network with perceptual mapping
	Black-box		
	Reed–Bore	[178, 179]	Neural exciter models
Neural	Flue-organ	[181–183]	Convolutional neural network
	Bowed-string	[180]	Adversarial autoencoder
	Pink Trombone	[184]	Transformer
	Guqin, plucked-strings...	[186–189]	Scattering recurrent networks
	Filtered delay loop	[193, 195]	
	White-box		
	Pink Trombone	[196]	Differentiable digital signal processing
	Scattering delay network	[197]	

SEC. 4.1.2) dataset and then use a Gaussian Mixture Model to learn a probabilistic mapping that predicts bow force and velocity [171]. Similarly, Smyth, Abel, and Smith invert a hybrid DWG-finite-difference time-domain avian vocal tract model (see SEC. 4.2) by precomputing the power spectra for different bronchial pressure and air volume flow values, used then with maximum-likelihood to estimate their trajectories from new recordings [143].

5.4 Genetic

Vuori and Välimäki first used genetic algorithms to optimize a flute model (see SEC. 4.1.1), defining a spectral fitness over fundamental and early harmonics [99]. Riionheimo and Välimäki extended this approach by introducing psychoacoustic weighting into the spectral fitness to optimize a plucked-string FDL [172]. Cooper et al. scaled this approach to a 2D-DWM vocal tract (see SEC. 4.2) tuning hundreds of spatial parameters [173]. Modern research by Cámara et al. on Pink Trombone optimization demonstrates genetic algorithms remain useful and can compete with black-box neural network approaches [174]. Furthermore, this strategy can be successfully hybridized; their recent work utilizes DDSP—see SEC. 5.5.2—to initialize parameters before refining the results with a genetic search [175].

5.5 Neural

5.5.1 Black-Box

Casey introduced the first neural approach to DWG optimization by inverting a simple bowed-string model through supervised training [176]. He fed DWG-synthesized audio y to a neural network and minimized the *parameter loss* $|\hat{\theta} - \theta|$, where $\hat{\theta}$ and θ are the predicted and known

DWG parameters, respectively, obtaining a *controller network* able to predict $\hat{\theta}$ from y . Additionally, he trained a second network as a *teacher* to predict output waveform $\hat{y}$ by minimizing $|\hat{y} - y|$ —which is referred to here as a *synthesizer network*—to help the controller network navigate nonconvex optimization landscapes.

Cemgil and Erkut furthered this idea by training a controller network for a plucked-string FDL, with an additional perceptual loss: a subjective mapping of θ obtained through listening tests [177]. Other work focused on using synthesizer networks to predict $e[n]$ for reed-bore DWGs, expanding the capabilities of DWGs through a data-driven nonlinearity [178, 179].

The introduction of modern neural network architectures has seen substantial improvements in the last decade. Sinclair trained an *adversarial autoencoder* on bowed-string (θ, y) pairs, obtaining a controllable synthesizer network that mirrors the behavior of the original DWG [180]. Gabrielli et al. demonstrated that *convolutional neural networks* achieve high-quality results when scaled to large datasets, culminating in a patented controller network for a flue-organ DWG [181–183]. Most recently, Xu and Reiss applied *transformers* to optimize area sections in the Pink Trombone vocal tract model, surpassing neural audio synthesis in reconstruction quality for vowel sounds [184].

5.5.2 White-Box

In white-box approaches, the DWG is embedded within the network, allowing backpropagation from output to parameters. Early research by Su, Liang, and collaborators exploited the equivalence between recurrent neural networks (RNNs) and IIR filters (see [185] for a mathematical deriva-

Table 2. Comparison of optimization methods.

Technique	DWG-Agnostic	Real Time	Data-Hungry	Quality
Physics-based	Yes~	No	No	Low
Filter design	No~	Yes	No	Medium
System identification	Yes	Yes	Yes	Medium
Genetic	Yes	No~	No	High
Black-box	Yes	Yes	Yes	High
White-box	Yes~	Yes	No~	High

Note. Nuance is indicated with “~” (discussed in the text).

tion). Hence, they designed RNNs whose structure mirrors DWG topologies, where neuron weights act as scattering coefficients [186–189]. However, RNN units need to be cascaded or parallelized to extend beyond first-order, hence complicating the topology of DWGs and impacting computational load at synthesis.

DDSP addresses this by implementing DWGs as native differentiable operations [188]. Parameters can then be optimized through a controller network or directly via gradient descent in a self-supervised manner. In the accompanying materials of Hayes et al. [199] an FDL filter’s decay is optimized by minimizing the spectral distance between target and synthesized audio at multiple resolutions [193], known as multiscale spectral loss ($\mathcal{L}_{MSS}$) [194]. Tablas de Paula et al. extend this approach to optimize pluck position, playing dynamics, timbre, and decay separately in a time-domain, time-variant implementation [195].

The same technique, gradient descent reducing $\mathcal{L}_{MSS}$, is applied to the Kelly-Lochbaum ladder-filter, where the vocal tract area sections are optimized, scoring higher results in perceptual listening tests than optimization with genetic algorithms, *particle-swarm optimization*, and Gabrielli’s black-box method from SEC. 5.5 [196]. Recently, DDSP techniques have been applied to SDNs (see SEC. 4.4) to match spectro-temporal features of a target room impulse response. The optimized SDN captures energy decay behavior more realistically than those modeled solely from geometry and tabulated wall absorption coefficients [197].

5.6 Comparisons and Current Challenges

In this review, physics-based and filter-design approaches have mostly targeted linear time-invariant FDLs. However, the former can target arbitrary DWG complexity given derivable or measurable parameters, while the latter supports precise frequency-dependent attenuation [198, 199]. System identification strategies remain useful for deriving input-output mappings, but their generalizability is limited. Genetic algorithms achieve high quality without data requirements, but reconstruction quality depends on population size, affecting real-time performance (Table 2). Modern approaches can significantly accelerate this process [175].

Once trained, neural approaches infer DWG parameters in real time and generalize to unseen data. Black-box

quality depends on data quality and size (*data-hungry*), while white-box methods mitigate this through synthesizer knowledge at the cost of requiring precise differentiable implementations. Both remain limited by loss functions and network capacity.

While parameter losses provide direct gradients, DDSP over-relies on $\mathcal{L}_{MSS}$. For sinusoidal frequency prediction, such as f_0 , $\mathcal{L}_{MSS}$ yields uninformative gradients [200–203]. To address this limitation, recent work explores alternative loss functions—including combinations of spectral, parameter, and discriminative objectives [204–206], along with hybridization with traditional heuristic searches to refine parameter initialization [175].

Subsequently, when *permutation symmetry* is present, point-based regressors will degrade performance, even when combined with permutation-invariant losses or symmetry-breaking heuristics [207]. Hayes et al. demonstrated that generative, flexible symmetry-equivariant architectures yield consistent gains in black-box controllers [207].

While generative prediction has entered DDSP, flexible symmetry-equivariant architectures remain absent [208]. In a related development, DDSP has opened a new direction: the combination of physically interpretable control of physical models with neural audio [209].

6 CONCLUSION

Over the past four decades, DWG applications have proliferated, as substantially covered in this review. It is a perfect example of how physically accurate simulations can be perturbed to reduce computations by orders of magnitude while preserving psychoacoustic equivalence. Just as the FM oscillator pair efficiently approximated a great number of oscillators in additive synthesis, the bidirectional delay lines and scattering junctions of DWG networks greatly reduced the burdensome finite-difference schemes of computational physics for wave propagation. The straightforward calculus of lossless wave scattering—often multiply-free—at sparse points within a general framework of bidirectional delay lines has been found to be intuitively appealing, physically meaningful, and computationally hyperefficient—frequently sounding equivalent to the ear while saving orders of magnitude in computation. Readers are encouraged to explore interactive demon-

strations of these principles in the Supplementary Material (<https://joshreiss.github.io/digital-waveguides-review/>).

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